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 Studierichting: Informatica
 Oefeningenreeks 5B nr. 2.6

$$\begin{aligned}
 & \int_{-\infty}^{+\infty} \frac{x}{x^2 + 4} dx \\
 &= \lim_{a \rightarrow -\infty} \int_a^0 \frac{x}{x^2 + 4} dx + \lim_{b \rightarrow +\infty} \int_0^b \frac{x}{x^2 + 4} dx \\
 & t = x^2 + 4 \left(dt = 2x dx, dx = \frac{dt}{2x}, x = \frac{1}{2} dt \right) \\
 &= \lim_{a \rightarrow -\infty} \int_a^0 \frac{1}{2} \frac{dt}{t} + \lim_{b \rightarrow +\infty} \int_0^b \frac{1}{2} \frac{dt}{t} \\
 &= \lim_{a \rightarrow -\infty} \frac{1}{2} [\ln |t|]_a^0 + \lim_{b \rightarrow +\infty} \frac{1}{2} [\ln |t|]_0^b \\
 & x^2 + 4 = t \\
 &= \lim_{a \rightarrow -\infty} \frac{1}{2} [\ln |x^2 + 4|]_a^0 + \lim_{b \rightarrow +\infty} \frac{1}{2} [\ln |x^2 + 4|]_0^b \\
 &= \lim_{a \rightarrow -\infty} \frac{1}{2} (\ln 4 - \ln(a^2 + 4)) + \lim_{b \rightarrow +\infty} \frac{1}{2} (\ln(b^2 + 4) - \ln 4) \\
 &= -\infty + +\infty, \text{ bestaat niet (onbepaalde vorm)}
 \end{aligned}$$

References